

RQ4.5 cell output

Notebook cell export: rq45_cell_output.pdf

Includes this cell's printed text, displayed tables, and generated figures.

Printed output

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[RQ4.5] Denominator: 4 loaded run(s) × 1000 tail episodes × 8 agents × 12 years = 384,000 expected cells (96,000 per run); actual counted cells = 384,000.
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Printed output

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[RQ4.5] Final non-compliant cells: 66,112 (17.2%); auction-stressed but compliant cells: 142,175 (37.0%).
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Printed output

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[RQ4.5] Non-compliant (agent, year, episode) cells in converged window: 66,112
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Printed output

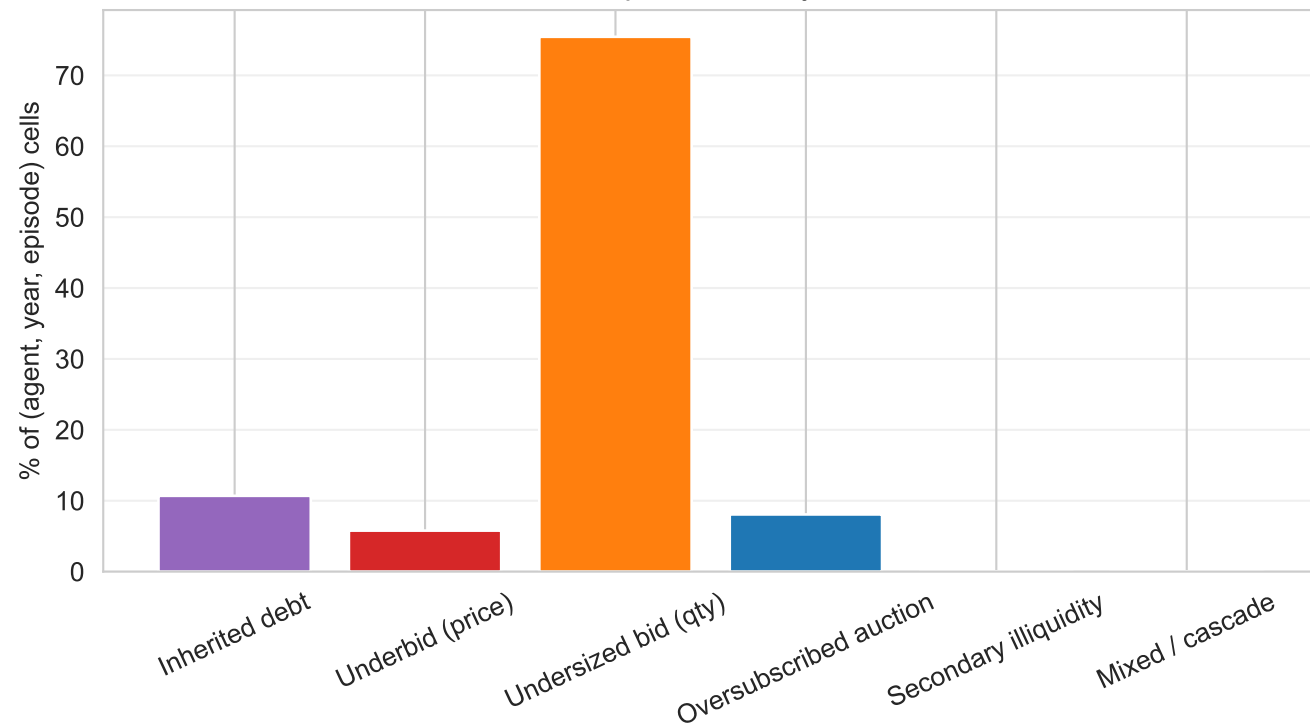
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[RQ4.5]  $\Sigma$  shortfall across the window: 79,149.41 Mt
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Cause counts and shares

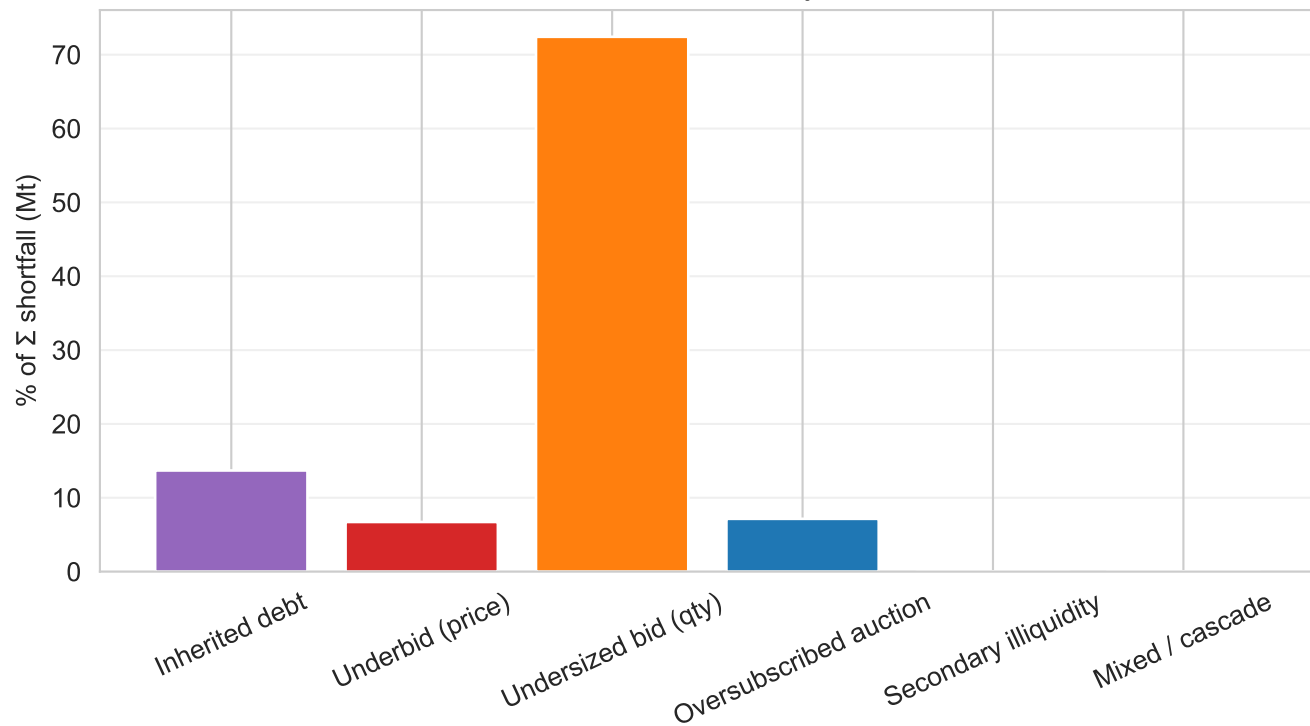
	n_cells	sum_shortfall_Mt	mean_shortfall_Mt	median_shortfall_Mt	cell_share_pct
tonne_share_pct					
cause					
Inherited debt 13.697	7070	10841.089	1.533	1.686	10.694
Underbid (price) 6.723	3825	5321.238	1.391	0.740	5.786
Undersized bid (qty) 72.424	49874	57323.488	1.149	1.099	75.439
Oversubscribed auction 7.151	5333	5659.925	1.061	0.718	8.067
Secondary illiquidity 0.003	4	2.079	0.520	0.547	0.006
Mixed / cascade 0.002	6	1.589	0.265	0.231	0.009

RQ4.5 — Why agents go non-compliant in the converged window (pooled across 4 loaded seeds)

Share of non-compliant cells by dominant cause



Share of total shortfall tonnes by dominant cause



Printed output

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[RQ4.5] Baseline and per-cause feature values (mean / median / std):
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Printed output

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All tail cells = normal/default-config bid baseline; Fully compliant cells = regular successful bid baseline.
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Baseline and per-cause feature values | columns 1-6 of 52

		n_cells	shortfall_Mt_mean	shortfall_Mt_median	shortfall_Mt_std
inherited_debt_Mt_mean	inherited_debt_Mt_median				
diagnostic_group					
All tail cells (normal/default)		384000	0.206	0.000	0.593
0.177	0.000				
Fully compliant cells		317888	0.000	0.000	0.000
0.060	0.000				
Inherited debt		7070	1.533	1.686	0.805
1.951	2.010				
Underbid (price)		3825	1.391	0.740	1.735
0.681	0.000				
Undersized bid (qty)		49874	1.149	1.099	0.800
0.611	0.078				
Oversubscribed auction		5333	1.061	0.718	1.177
0.370	0.000				
Secondary illiquidity		4	0.520	0.547	0.362
0.157	0.049				
Mixed / cascade		6	0.265	0.231	0.180
0.030	0.000				

Baseline and per-cause feature values | columns 7-12 of 52

		inherited_debt_Mt_std	alloc_Mt_mean	alloc_Mt_median	alloc_Mt_std
emissions_Mt_mean	emissions_Mt_median				
diagnostic_group					
All tail cells (normal/default)		0.543	1.739	1.563	1.217
1.765	1.675				
Fully compliant cells		0.295	1.797	1.582	1.282
1.647	1.525				
Inherited debt		0.809	2.250	2.260	0.702
1.938	1.986				
Underbid (price)		1.429	0.000	0.000	0.000
1.519	1.535				
Undersized bid (qty)		0.797	1.556	1.541	0.586
2.498	2.596				
Oversubscribed auction		0.894	0.509	0.270	0.609
1.847	1.807				
Secondary illiquidity		0.252	1.614	1.826	0.896
1.976	1.994				
Mixed / cascade		0.075	1.133	1.096	0.542
1.368	1.327				

Baseline and per-cause feature values | columns 13-18 of 52

		emissions_Mt_std	bid_price_mean	bid_price_median	bid_price_std
clearing_price_mean	clearing_price_median				
diagnostic_group					
All tail cells (normal/default)		0.868	96.857	91.032	46.931
54.383	45.000				
Fully compliant cells		0.835	91.997	84.665	44.816
53.839	45.000				
Inherited debt		0.584	148.542	165.621	44.188
69.600	61.282				
Underbid (price)		0.584	61.477	45.000	26.881
98.740	94.685				
Undersized bid (qty)		0.781	128.014	135.097	44.419
52.539	45.000				
Oversubscribed auction		0.614	52.076	45.000	19.354
52.077	45.000				
Secondary illiquidity		0.667	45.000	45.000	0.000
45.000	45.000				
Mixed / cascade		0.503	45.000	45.000	0.000
45.000	45.000				

Baseline and per-cause feature values | columns 19-24 of 52

		clearing_price_std	bid_to_clearing_mean	bid_to_clearing_median	bid_to_clearing_std
bid_qty_mult_mean	bid_qty_mult_median				
diagnostic_group					
All tail cells (normal/default)		20.575	1.874	1.671	0.934
1.298	1.433				
Fully compliant cells		19.689	1.789	1.545	0.891
1.418	1.814				
Inherited debt		28.262	2.329	2.226	0.906
0.676	0.500				
Underbid (price)		33.604	0.647	0.639	0.201
1.731	2.000				
Undersized bid (qty)		18.948	2.537	2.535	0.911
0.522	0.500				
Oversubscribed auction		19.357	1.000	1.000	0.000
1.890	2.000				
Secondary illiquidity		0.000	1.000	1.000	0.000
1.097	1.111				
Mixed / cascade		0.000	1.000	1.000	0.000
1.356	1.214				

Baseline and per-cause feature values | columns 25-30 of 52

		bid_qty_mult_std	bid_coverage_ratio_mean	bid_coverage_ratio_median
bid_coverage_ratio_std	estimate_need_Mt_mean	estimate_need_Mt_median		
diagnostic_group				
All tail cells (normal/default)		0.681	1.346	1.403
0.771	1.971	1.742		
Fully compliant cells		0.652	1.476	1.657
0.752	1.737	1.549		
Inherited debt		0.424	0.713	0.553
0.416	4.179	4.365		
Underbid (price)		0.465	1.781	1.935
0.554	2.247	1.835		
Undersized bid (qty)		0.109	0.517	0.501
0.081	3.100	3.085		
Oversubscribed auction		0.286	1.885	1.933
0.363	2.190	1.957		
Secondary illiquidity		0.084	1.011	1.019
0.133	1.985	2.036		
Mixed / cascade		0.530	1.106	1.141
0.166	1.301	1.309		

Baseline and per-cause feature values | columns 31-36 of 52

		estimate_need_Mt_std	need_realised_Mt_mean	need_realised_Mt_median
need_realised_Mt_std	undersized_qty_Mt_mean	undersized_qty_Mt_median		
diagnostic_group				
All tail cells (normal/default)		1.139	1.942	1.726
1.129	0.509	0.000		
Fully compliant cells		0.943	1.708	1.540
0.936	0.338	0.000		
Inherited debt		1.375	3.889	4.000
1.276	1.539	1.733		
Underbid (price)		1.557	2.200	1.818
1.526	0.081	0.000		
Undersized bid (qty)		1.182	3.109	3.089
1.194	1.540	1.543		
Oversubscribed auction		1.127	2.217	1.986
1.141	0.017	0.000		
Secondary illiquidity		0.836	2.134	2.036
0.887	0.125	0.049		
Mixed / cascade		0.657	1.398	1.327
0.522	0.030	0.000		

Baseline and per-cause feature values | columns 37-42 of 52

	undersized_qty_Mt_std	oversubscription_ratio_mean	oversubscription_ratio_median
oversubscription_ratio_std	bank_start_Mt_mean	bank_start_Mt_median	
diagnostic_group			
All tail cells (normal/default)	0.745	1.275	1.247
0.332	2.017	1.595	
Fully compliant cells	0.599	1.260	1.234
0.314	2.372	1.956	
Inherited debt	0.853	1.543	1.538
0.346	0.000	0.000	
Underbid (price)	0.323	1.938	1.746
0.646	0.624	0.317	
Undersized bid (qty)	0.658	1.258	1.234
0.329	0.303	0.000	
Oversubscribed auction	0.138	1.474	1.423
0.305	0.552	0.226	
Secondary illiquidity	0.191	1.036	1.036
0.008	0.000	0.000	
Mixed / cascade	0.074	1.033	1.037
0.014	0.000	0.000	

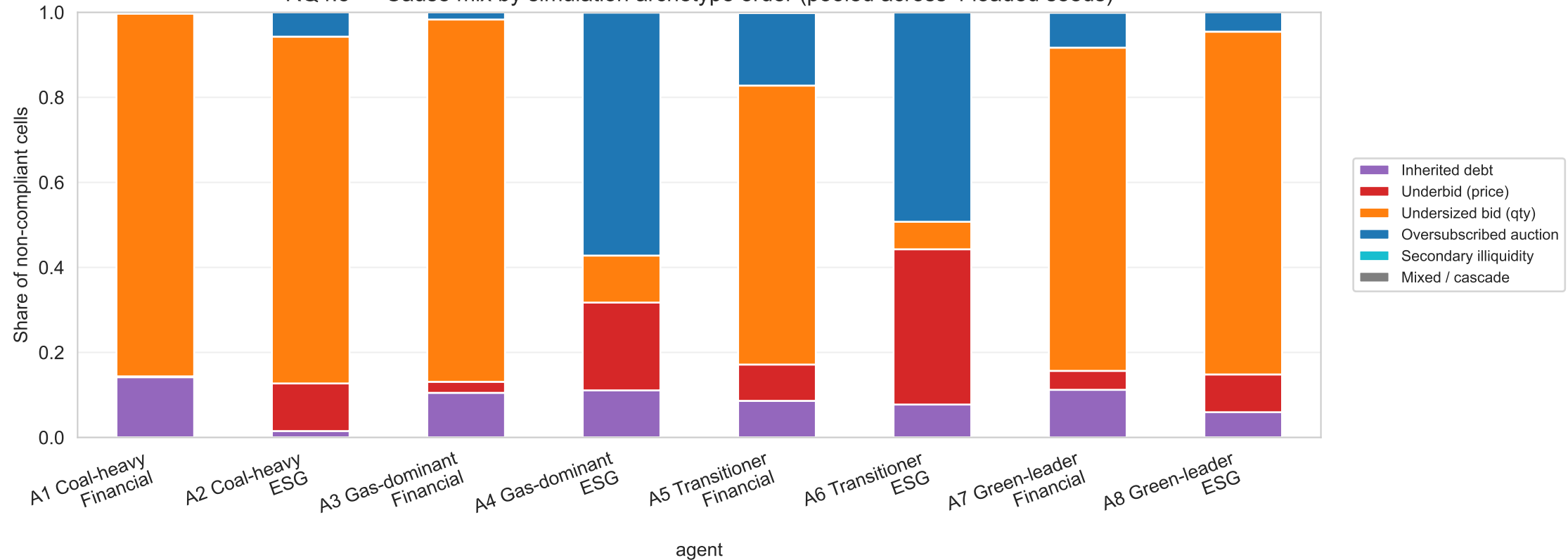
Baseline and per-cause feature values | columns 43-48 of 52

		bank_start_Mt_std	sec_buy_intent_Mt_mean		sec_buy_intent_Mt_median
sec_buy_intent_Mt_std	secondary_buy_unmet_Mt_mean	secondary_buy_unmet_Mt_median			
diagnostic_group					
All tail cells (normal/default)		1.991		1.669	3.000
1.474	1.301		0.0		
Fully compliant cells		2.000		1.444	0.071
1.482	1.021		0.0		
Inherited debt		0.000		2.768	3.000
0.786	2.663		3.0		
Underbid (price)		0.713		2.407	3.000
1.176	2.223		3.0		
Undersized bid (qty)		0.498		2.815	3.000
0.706	2.714		3.0		
Oversubscribed auction		0.667		2.406	3.000
1.175	2.312		3.0		
Secondary illiquidity		0.000		3.000	3.000
0.000	3.000		3.0		
Mixed / cascade		0.000		0.000	0.000
0.000	0.000		0.0		

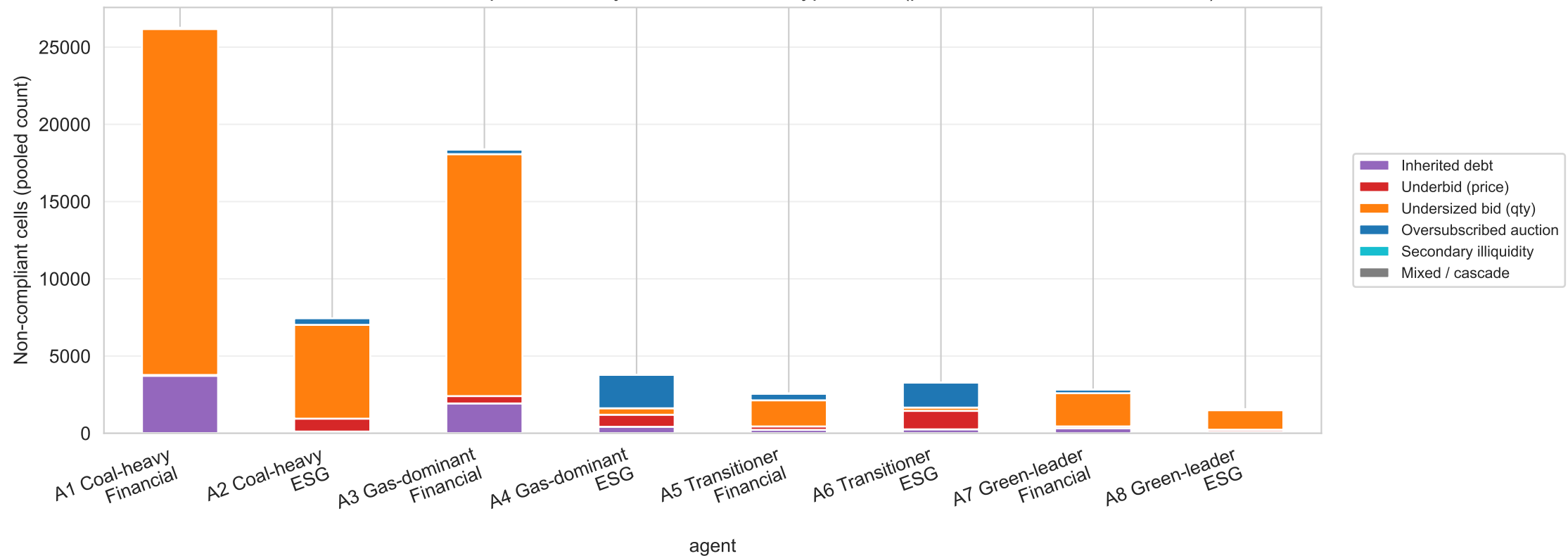
Baseline and per-cause feature values | columns 49-52 of 52

	secondary_buy_unmet_Mt_std	loan_outstanding_Meur_mean	loan_outstanding_Meur_median
loan_outstanding_Meur_std			
diagnostic_group			
All tail cells (normal/default)	1.449	0.0	0.0
0.0			
Fully compliant cells	1.384	0.0	0.0
0.0			
Inherited debt	0.864	0.0	0.0
0.0			
Underbid (price)	1.276	0.0	0.0
0.0			
Undersized bid (qty)	0.797	0.0	0.0
0.0			
Oversubscribed auction	1.233	0.0	0.0
0.0			
Secondary illiquidity	0.000	0.0	0.0
0.0			
Mixed / cascade	0.000	0.0	0.0
0.0			

RQ4.5 — Cause mix by simulation archetype order (pooled across 4 loaded seeds)



RQ4.5 — Absolute non-compliant cells by simulation archetype order (pooled across 4 loaded seeds)



Printed output

[RQ4.5] Compliant vs non-compliant cell means and medians (key discriminators):

Compliant vs non-compliant feature contrast | columns 1-6 of 7

	feature	compliant_mean	non_compliant_mean	delta_mean_(nc - c)	compliant_median	non_compliant_median
0	inherited_debt_Mt	0.060	0.739	0.678	0.000	0.173
1	bank_start_Mt	2.372	0.309	-2.063	1.956	0.000
2	bid_qty_mult	1.418	0.719	-0.700	1.814	0.500
3	bid_to_clearing	1.789	2.281	0.492	1.545	2.271
4	coverage_gap_Mt	0.481	2.104	1.623	0.000	2.107

Compliant vs non-compliant feature contrast | columns 7-7 of 7

	delta_median_(nc - c)
0	0.173
1	-1.956
2	-1.314
3	0.726
4	2.107

Printed output

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[RQ4.5] Auction-stressed but ultimately compliant cells: 142,175
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Printed output

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Stressed = auction allocation < emissions + carry-forward start; final shortfall = 0.
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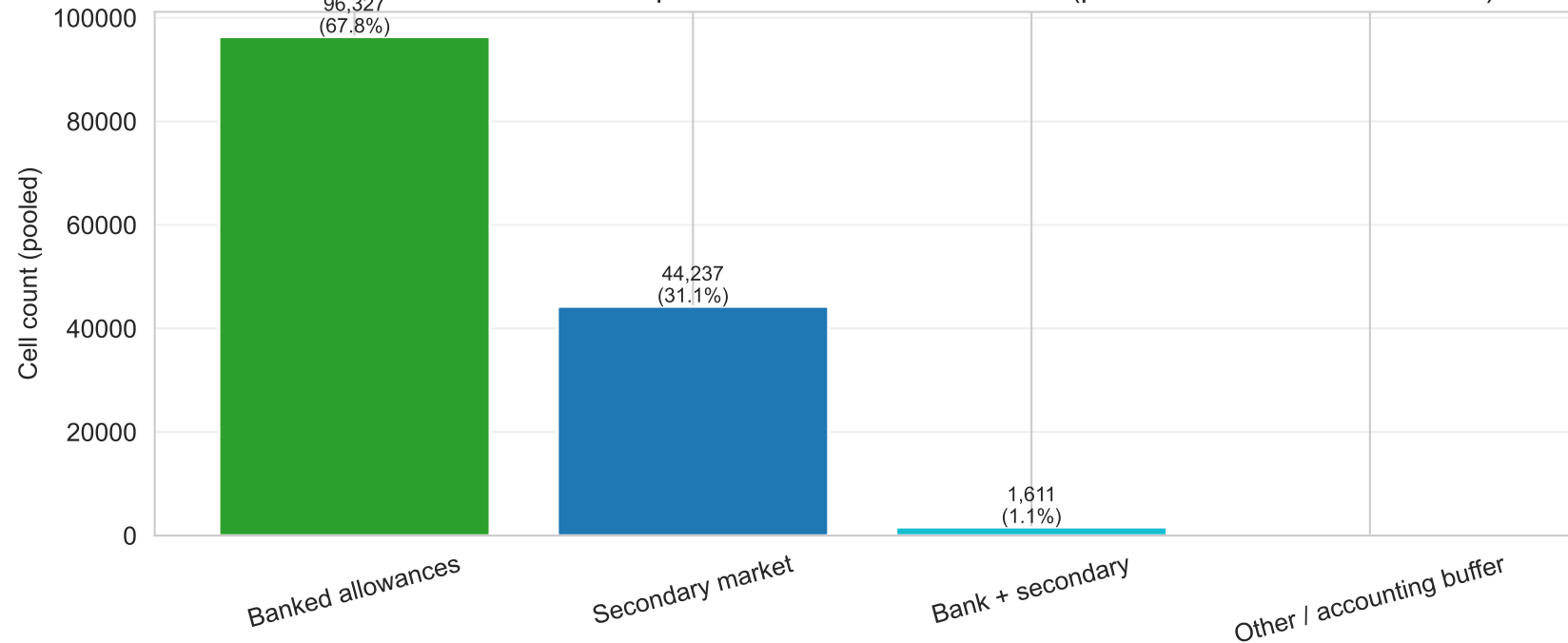
Auction-stressed compliant cells by solution | columns 1-6 of 12

	n_cells	share_pct	mean_initial_gap_Mt	median_initial_gap_Mt	sum_initial_gap_Mt
mean_bank_start_Mt					
solution					
Banked allowances	96327	67.752	0.890	0.840	85738.173
3.020					
Secondary market	44237	31.114	1.650	1.627	72984.968
0.937					
Bank + secondary	1611	1.133	1.932	1.722	3113.214
1.228					
Other / accounting buffer	0	0.000	0.000	0.000	0.000
0.000					

Auction-stressed compliant cells by solution | columns 7-12 of 12

		mean_bank_used_estimate_Mt	mean_secondary_buy_Mt	sum_secondary_buy_Mt
mean_secondary_paid_Meur	sum_secondary_paid_Meur	mean_secondary_paid_price_EUR_t		
solution				
Banked allowances		0.870	0.020	1948.997
1.849	1.780747e+05		89.952	
Secondary market		0.000	2.837	125519.232
410.794	1.817231e+07		142.228	
Bank + secondary		0.478	1.455	2343.612
215.366	3.469541e+05		127.979	
Other / accounting buffer		0.000	0.000	0.000
0.000	0.0000000e+00		0.000	

RQ4.5 — Auction-stressed compliant cells: resolution channel (pooled across 4 loaded seeds)



Printed output

```
[RQ4.5] Writing cell output PDF to: C:\Users\danie\Documents\Python_Scripts\Master_Thesis\Thesis-Energy-  
Auction\results\analysis\rq45_cell_output.pdf
```